

The Triangular Balance of Environment, Economy, and Society: A System Dynamics-Based Model for Sustainable Tourism Management

Summary

As global tourism continues to thrive, the issue of "**overtourism**" has become increasingly pronounced. Taking Juneau, Alaska, as a case study, the overwhelming influx of tourists poses significant challenges to local governments, including overstressed infrastructure, accelerated environmental degradation, and declining community satisfaction. Balancing the **triangular relationship** between **economic, social, and environmental sustainability** while meeting tourist demands has become a central focus for academics and policymakers worldwide. This paper explores various **policies and management strategies** that locations can adopt to achieve sustainable tourism development and proposes an **optimized solution** for governmental implementation.

To address these challenges, we developed an **improved system dynamics model** to quantify the interactions among the environmental, social, and economic dimensions. This model systematically evaluates four **interdependent variables: tourist numbers, environmental degradation index, infrastructure capacity, and resident satisfaction**. Additionally, it incorporates **tax revenue** as an **intermediary variable**, establishing a **dynamic feedback mechanism**. By adjusting tax allocation weights, tax rates, and policy-driven fees, we simulate the impacts of **policy changes** on tourism sustainability and identify optimal strategies. Using the **L-BFGS-B algorithm**, we determine the policy combinations that maximize social welfare. The model integrates dynamic parameters derived from **multi-source data and real-world case studies**, ensuring its broad applicability. Through sensitivity analysis and scenario simulations, we reveal the critical importance of **policy prioritization and phased adjustments** in achieving sustainability goals. Furthermore, the model's adaptability and generalizability were validated by extending its application to other regions, such as Sanya, China. As an innovative contribution, we propose a strategy to promote the development of low-traffic tourist destinations, exploring opportunities for more balanced tourism growth.

This study provides scientific evidence for sustainable tourism management in Juneau and offers theoretical insights and practical tools to address overtourism in similar global destinations. By proposing innovative strategies, the research aims to contribute to solving the challenges of overtourism and advancing tourism towards a new era of environmental sustainability and social harmony. In some cases, it can also help with the development of some undervisited destinations.

Keywords:

System Dynamics; differential equations; logistics growth model; dynamic feedback loops; L-BFGS-B algorithm; sustainable tourism; dynamic regulation; environmental protection; social welfare; economic balance